

Goldman Sachs Info (for SDE role)

14 questions will be there , 3 coding questions . Don't worry won't be tough (one easy , leetcode medium - hard level) , rest mcq s on aptitude , PnC , Code Snippet output , Probability , DS Algorithms mainly . As per the trend of Hacker Rank , there is a high chance of exam is repeated , Toh rat ke jaana kaske . Agar apne kisi dost ki placement lagwani ho toh use bhi bhejdiyo (bhai chutiya hai kya , mat bhejiyo baad mein regret karega!!!

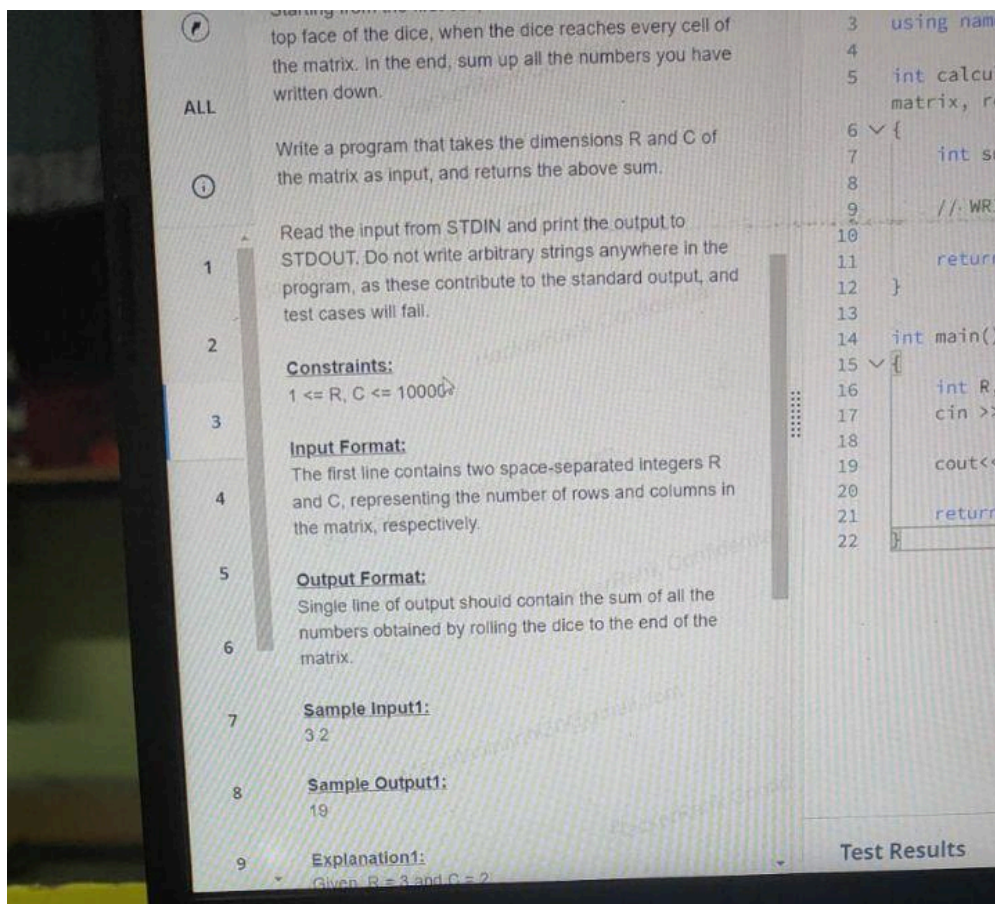
Chal padhle , All the best !!

I could recall following stuff and have some photographs too :

Coding -

- 1) A powerful string is a kind of string in which all the characters are same.
You are given a string s of length n and you are asked to calculate the total number of ways in which the given string s can be made powerful string by removing exactly one substring from the given input string. Print the ans module 10^9+7

2)



- 3) Third problem was somewhat based on color , means we are given 1*3 arrays of pure colors and some colors in an array in binary format . You've to find the pure color which are closest to given color :

I'm attaching java code which I got from somewhere .

```
public class Main{
    public static void main(String[] args){
        String test = "010111011010010110000011" ;
        System.out.println(test.length());
        int[][] paure = {{0,0,0},{255,255,255},{255,0,0},{0,255,0},{0,0,255}};
        String[] pure = {"Black" , "White" , "Red" , "Green" , "Blue" } ;
        int n1 = Integer.parseInt(test.substring(0,8),2) ;
        int n2 = Integer.parseInt(test.substring(8,16),2) ;
        int n3 = Integer.parseInt(test.substring(16,24),2) ;
        System.out.println(n1+" "+n2+" "+n3);
        int mindist = Integer.MAX_VALUE , color = -1 ;
        for(int i=0;i<5;i++){
            int[] base = paure[i] ;
            int dis = Math.abs(base[0]-n1) + Math.abs(base[1]-n2) +
Math.abs(base[2]-n3) ;
            if(dis<mindist){
                color = i;
                mindist = dis;
            }
            else if(dis==mindist){
                System.out.println("Ambiguous");
            }
        }

        System.out.print(pure[color]) ;
    }
}
```

MCQs:

1)

12. Question 12

7 people visited a marriage function in 3 different vehicles, each accommodating a maximum of 5 persons. In how many ways can they visit, such that they use all the 3 vehicles?

Pick **ONE** option

☐ 3600

☐ 1500

☐ 1650

☐ 1806

Year Selection

2)

9. Question 9

Given that $\log_3(p(\log_2(q + \log_3(1 + 2\log_2 x)))) = 1/2$, 'p' is the positive root of the quadratic equation: $x^2 + 998x - 3003 = 0$ and 'q' is the least value of the polynomial $(x^3 - 2x + 2)$. Find the value of x.

Pick **ONE** option

☐ 2

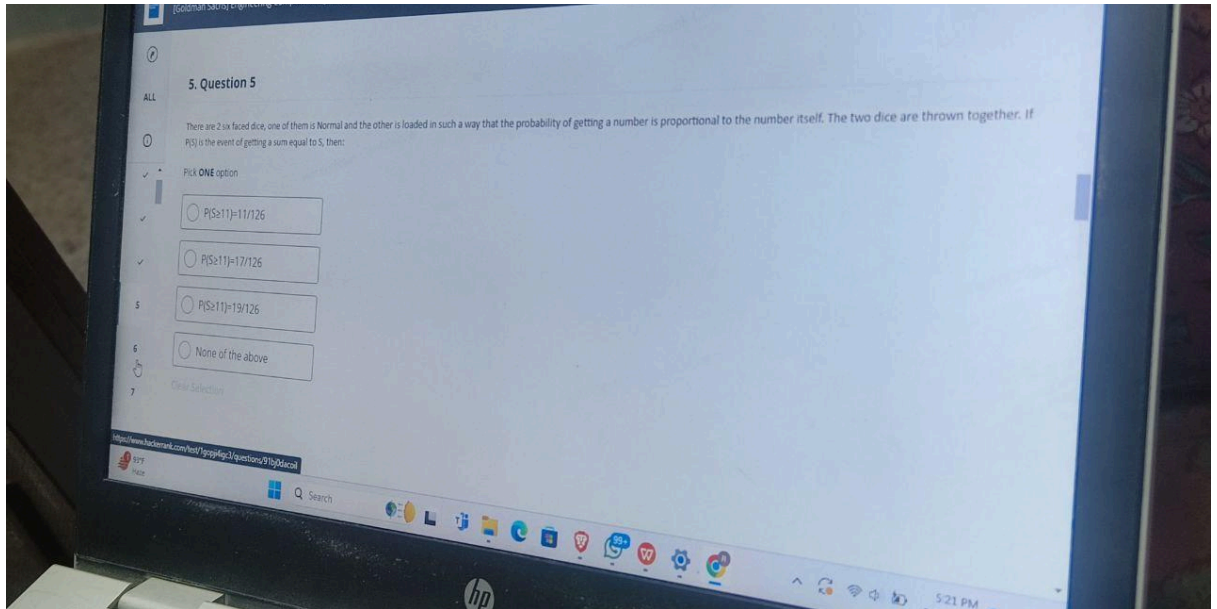
☐ 4

☐ 3

☐ 5

Year Selection

- 3) Form of code that uses more than one process and processor and may on occasions have more than one process or processor active at the same time, is called : I was confused between **Mutiprocessing** or **Multi-threading** as answer .



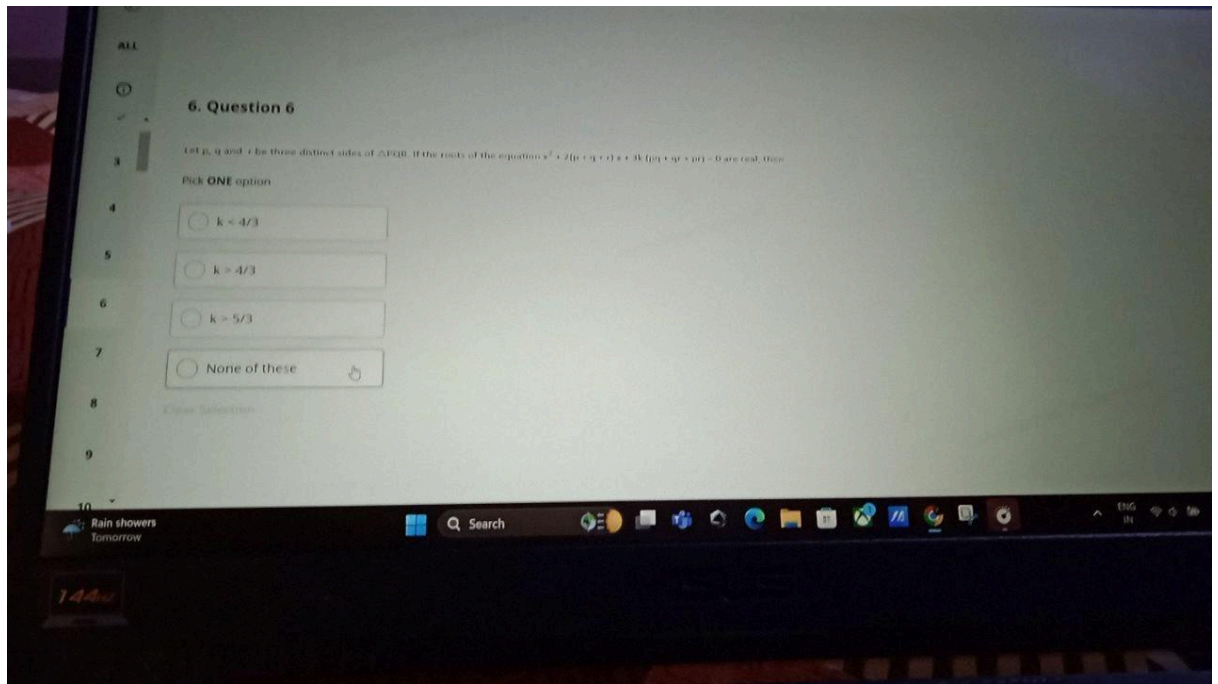
4)

- 5) A question on **Linear Hashing** :

Just for your knowledge - Linear hashing is a type of dynamic hashing that helps address the issue of overflow chains by dynamically resizing the hash table and splitting buckets when they overflow.

- 6) Which of the following can be defined as private : Constructor , static methods , Abstract Methods –

Ans in my opinion - Only (i) and (ii) can be private , but Abstract methods can not be



7)

- 8) Question based on address of Aij , if base addresses , array size and block size are given

As per google

In a row-major matrix :

Address of A[i][j]=Base

Address+((j-col_start)+(i-row_start)×num_columns)×element_size

- 9) 20 students in class , half boys refused for handshake with one-third of the girls , find possible number of handshakes . Answer may be 190 probably

Bas itna hi yaad hai bhai .